Affective visualization graphs

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Table of contents

Setting up

```
# read in packages
library(tidyverse) # general use
library(here) # file organization -- dont need unless sort into folders
library(janitor) # cleaning data frames
```

```r
library(DHARMa) # for diagnostics
library(ggeffects) # for ggpredict
library(flextable) # for flextable
library(gtsummary)
```

```r
# read in personal data to object from csv
my_data <- read_csv("envs193ds-personal-data.csv")
```

Framing my question

My central question is: How does sleep duration the night prior influence the probability of me going on a run the next day?

Variables

- Predictor: sleep duration the previous night (hours)
- Response: probability of going on a run (1/0) (1 = yes; 0 = no run)

Cleaning and wrangling data

```r
# cleaned column names
my_data <- my_data |>
  clean_names()

# created new column: 1 if workout was a run, 0 if not
my_data <- my_data |>
  mutate(is_run = case_when(
    workout_type == "Run" ~ 1, # run = 1
    .default = 0))             # all other workout types = 0
```

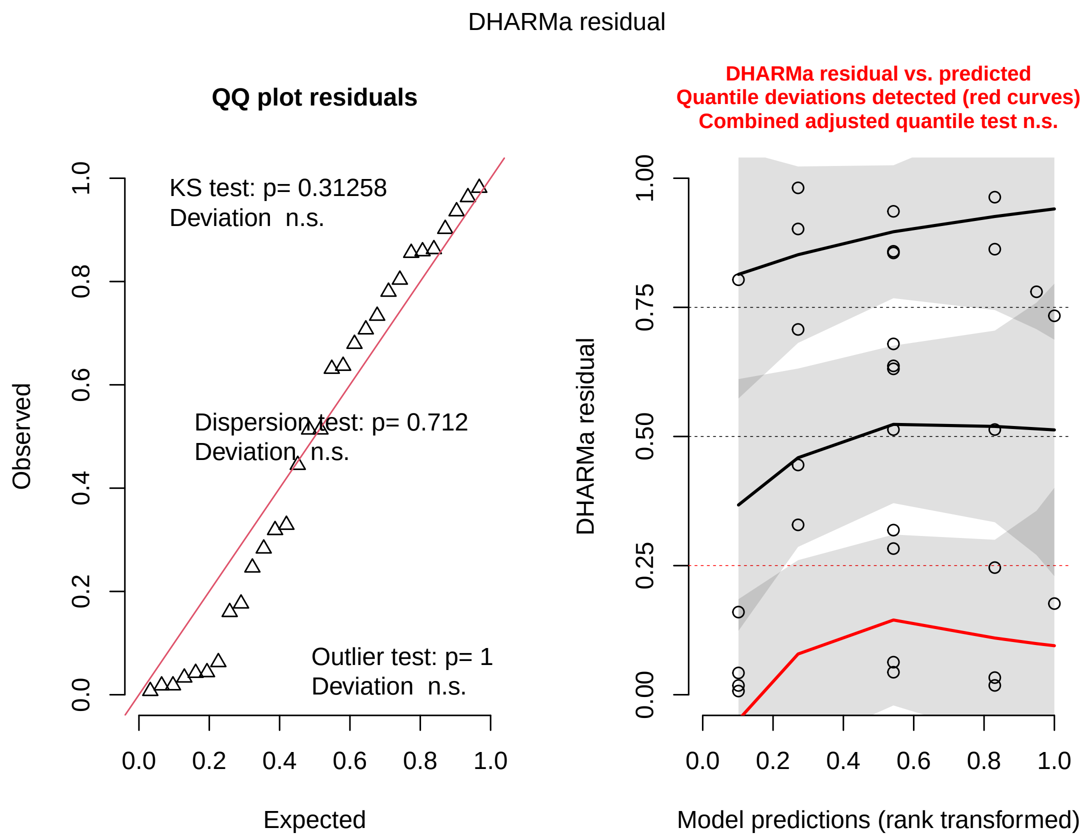
Fitting the model

```r
# fitted generalized linear model for running
# response: whether or not workout was a run (binary: 1/0)
# predictor: sleep the previous night (hours)
run_mod <- glm(
```

```
  # formula: binary run outcome ~ sleep hours
  is_run ~ sleep_hours,
  # data: my_data data frame
  data = my_data,
  # binomial error distribution for binary response
  family = binomial)
```

Checking diagnostics

```
# simulated residuals from GLM for diagnostic plots
plot(simulateResiduals(run_mod))
```



Info from left plot (QQ plot of residuals):

- KS test: do my residuals follow a uniform distribution? → generally yes (CHECK)

- Dispersion test: are your residuals more dispersed than expected from a uniform distribution? → no
- Outlier test: are there outliers in your data set that could be influencing model coefficients or predictions? → no

Info from right plot (residuals vs predicted values):

- By default, R generates the reference lines for quartiles: 25, 50, 75
- Not that informative but look at patterns within residuals across the range → DHARMa diagnostics flag issues in 25th quartile due to small sample size, not because the model is wrong.

Looking at the model coefficients

```r
# displayed model summary
summary(run_mod)
```

```

Call:
glm(formula = is_run ~ sleep_hours, family = binomial, data = my_data)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)  
(Intercept)  -8.7011     3.4964  -2.489   0.0128 *
sleep_hours   1.0873     0.4712   2.307   0.0210 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 36.652  on 29  degrees of freedom
Residual deviance: 28.046  on 28  degrees of freedom
AIC: 32.046

Number of Fisher Scoring iterations: 5
```

Calculate inverse logit

Characteristic	OR	95% CI	p-value
sleep_hours	2.97	1.37, 9.40	0.021

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```r
# calculated inverse logit
plogis(-8.7011)
```

```
[1] 0.000166375
```

Note:

- inverse logit is 0.00016
- The probability of going on a run if sleep duration is 0 hours is 0.00016??

Calculate odds ratio

```r
# calculated odds ratio
gtsummary::tbl_regression(run_mod,
                          exponentiate = TRUE)
```

Note:

- results at top of this page
- odds ratio is 2.97
- 2.97 > 1 so the probability of going on a run increases by a factor of 2.97 with every 1 hour increase in sleep

Summary: For every 1 hour increase in sleep, the the odds of going on a run increase by a factor of 2.97 (95% CI: [1.37, 9.40], $p = 0.021$, $= 0.05$).

Other way to calculate – remove?

```r
# exponentiated coefficients to get odds ratios
# e.g. exp(coef) = odds ratio for 1 hr increase in sleep
exp(coef(run_mod))
```

```
 (Intercept)  sleep_hours
0.0001663954 2.9663093920
```

Looking at the model predictions

- using probabilities to determine chance of getting certain outcome
- line represents the probability of 1 (going on a run) actually occurring

```r
# what is the probabilty of going on a run after sleeping for 10 hours?
ggpredict(run_mod,
          terms = "sleep_hours [10]")
```

```
# Predicted probabilities of is_run

sleep_hours | Predicted |     95% CI
------------------------------------
         10 |      0.90 | 0.39, 0.99
```

```r
# what is the probabilty of going on a run after sleeping for 4 hours?
ggpredict(run_mod,
          terms = "sleep_hours [4]")
```

```
# Predicted probabilities of is_run

sleep_hours | Predicted |     95% CI
------------------------------------
          4 |      0.01 | 0.00, 0.25
```

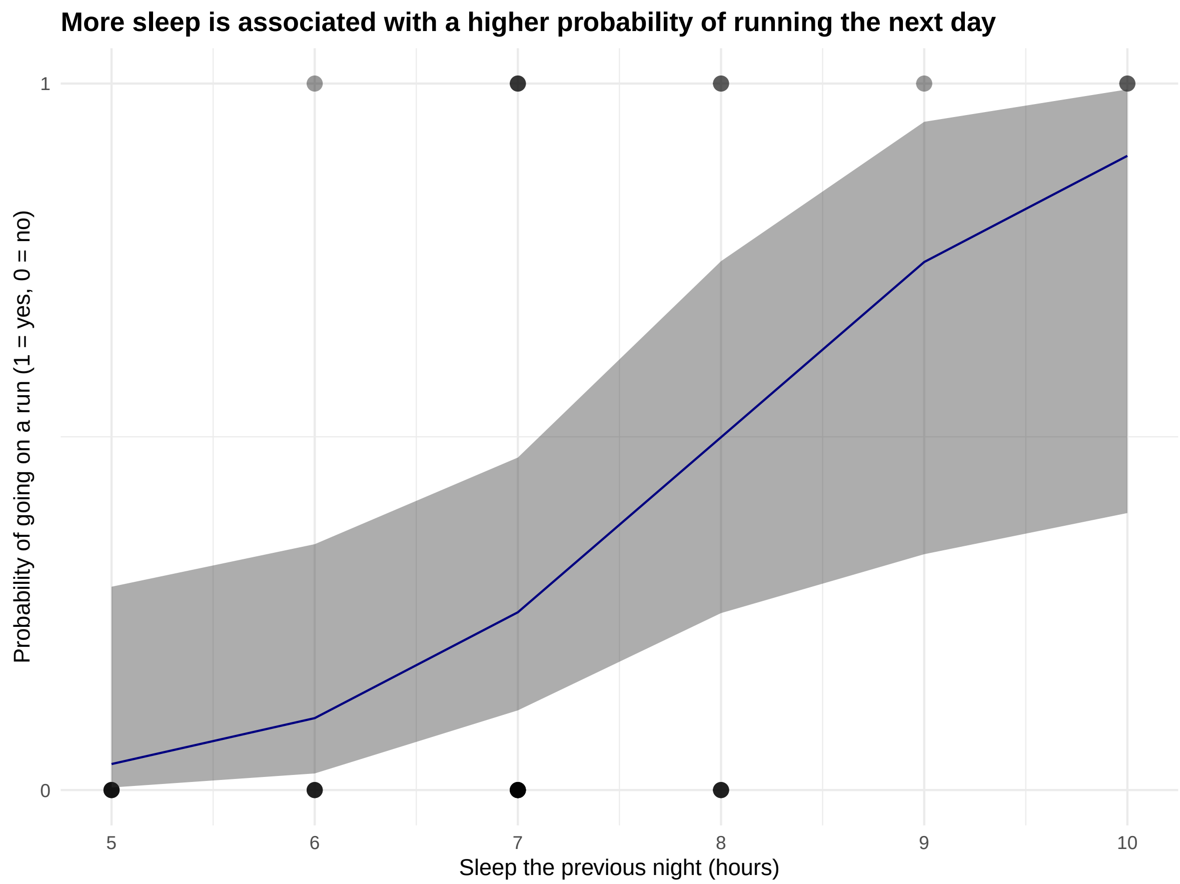
Note:

- If I slept for 10 hours, the probability of going on a run the next day is 0.90 (95% CI [0.39, 0.99]).
- If I slept for 4 hours, the probability of going on a run the next day is 0.01 (95% CI [0.00, 0.25]).

Visualizing model predictions

```r
# generated model predictions across range of sleep values in data set
mod_preds <- ggpredict(run_mod,
                       terms = "sleep_hours [5:10 by = 1]")

# base layer: ggplot with raw data
# data frame: my_data
ggplot(my_data,
       aes(x = sleep_hours,
           y = is_run)) +
  # first layer: raw observations
  geom_point(size = 3,
             alpha = 0.4) +
  # second layer: confidence interval ribbon from model predictions
  geom_ribbon(data = mod_preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high),
                  alpha = 0.4) +
  # third layer: logistic curve from model predictions
  geom_line(data = mod_preds,
            aes(x = x,
                y = predicted),
            # custom color
            color = "navyblue") +
  # constrained y-axis to 0 and 1
  scale_y_continuous(limits = c(0, 1),
                     breaks = c(0, 1)) +
  # relabeled axes and added title
  labs(title = "More sleep is associated with a higher probability of running the next day",
       x     = "Sleep the previous night (hours)",
       y     = "Probability of going on a run (1 = yes, 0 = no)") +
  # changed theme from default
  theme_minimal() +
  # bolded title
  theme(plot.title = element_text(face = "bold"))
```



Describing the model

```
# displaying model in table
as_flextable(run_mod)
```

	Estimate	Standard Error	z value	Pr(>|z|)	
(Intercept)	-8.701	3.496	-2.489	0.0128	*
sleep_hours	1.087	0.471	2.307	0.0210	*

	Estimate	Standard Error	z value	Pr(>|z|)

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 36.65 on 29 degrees of freedom

Residual deviance: 28.05 on 28 degrees of freedom